

PETROVSKIY ZAVOD, Russian Federation

WMO: 308380

Lat: 51.317N

Lon: 108.867E

Elev: 806

StdP: 92.01

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -38.8	(c) -36.5	(d) -40.6	(e) 0.1	(f) -38.7	(g) -38.4	(h) 0.1	(i) -36.4	(j) 6.5	(k) -16.3	(l) 5.8	(m) -17.0	(n) 0.3	(o) 10	(p) 0.670

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	29.4	16.7	27.5	16.2	25.6	15.5	18.5	25.7	17.7	24.2	16.9	23.1	2.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	12.8	20.4	15.5	12.1	19.8	14.6	11.4	19.4	56.2	25.8	53.3	24.3	50.8	23.1	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.3	(b) 5.5	(c) 4.8	(e) -42.0	(f) 33.4	(g) 3.0	(h) 2.1	(i) -44.2	(j) 34.9	(k) -46.0	(l) 36.1	(m) -47.7	(n) 37.2	(o) -49.9	(p) 38.7		
(a) 6.3	(b) 5.5	(c) 4.8	(e) -41.8	(f) 20.2	(g) 2.9	(h) 1.0	(i) -43.9	(j) 20.9	(k) -45.6	(l) 21.4	(m) -47.3	(n) 22.0	(o) -49.4	(p) 22.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.2	-24.2	-19.8	-9.5	0.8	7.2	14.6	17.5	14.7	7.6	-1.6	-13.1	-21.4		
	DBStd	15.22	5.36	5.70	6.65	4.62	3.99	3.93	3.05	3.42	4.01	4.92	6.35	6.17		
	HDD10.0	5009	1061	834	604	277	105	7	0	6	93	358	692	972		
	HDD18.3	7523	1319	1068	863	526	344	122	52	118	322	617	942	1231		
	CDD10.0	568	0	0	0	1	19	144	232	151	21	0	0	0		
	CDD18.3	41	0	0	0	0	1	9	25	6	0	0	0	0		
	CDH23.3	971	0	0	0	1	36	303	466	147	19	0	0	0		
	CDH26.7	259	0	0	0	0	7	79	145	26	2	0	0	0		
	WSAvg	1.9	1.4	1.4	2.0	2.4	2.6	2.3	2.0	1.9	2.0	1.7	1.6	1.4		
	PrecAvg	328	4	4	5	11	26	48	75	84	45	12	9	7		
	PrecMax	503	11	8	21	36	94	108	142	157	103	31	14	13		
(15) Precipitation	PrecMin	261	0	1	0	1	6	8	33	28	16	0	3	1		
	PrecStd	55	3	2	4	9	19	23	33	34	22	7	3	3		
	DB	-7.6	0.1	10.5	20.7	27.6	32.1	33.0	29.6	25.9	17.3	4.7	-4.8			
	MCWB	-8.7	-3.3	3.1	8.3	12.8	16.6	19.1	17.4	14.2	7.9	0.6	-6.3			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	-11.4	-4.5	6.8	16.4	23.7	28.7	30.1	26.9	22.2	13.0	1.4	-7.5			
	2% MCWB	-12.4	-6.5	1.0	6.6	11.2	15.7	17.7	16.5	12.6	5.7	-1.5	-8.5			
	5% DB	-13.6	-7.1	4.1	13.3	20.8	26.8	28.0	24.8	19.2	9.8	-0.9	-9.7			
	5% MCWB	-14.4	-8.8	-0.7	4.8	9.6	15.3	17.1	16.2	11.3	4.0	-3.0	-10.6			
	10% DB	-15.6	-9.5	1.2	10.4	17.8	24.7	25.9	22.6	16.4	6.7	-3.0	-11.6			
	10% MCWB	-16.2	-10.8	-2.3	3.4	8.0	14.6	16.6	15.4	10.1	2.4	-4.6	-12.2			
	0.4% WB	-8.9	-3.1	3.3	8.8	13.3	18.5	20.2	19.1	15.3	8.4	0.8	-6.1			
	0.4% MCDB	-7.7	-0.3	9.6	18.8	25.9	27.0	28.3	25.0	23.3	17.0	4.1	-5.0			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	-12.3	-6.3	1.2	6.9	11.7	17.4	19.0	17.8	13.6	6.2	-1.2	-8.4			
	2% MCDB	-11.5	-4.4	6.4	15.5	22.3	25.1	26.8	23.4	20.1	12.5	1.2	-7.6			
	5% WB	-14.4	-8.7	-0.5	5.1	10.1	16.4	18.2	17.1	12.2	4.3	-2.9	-10.5			
	5% MCDB	-13.6	-7.1	3.7	12.7	19.4	23.8	25.1	22.4	17.7	9.0	-1.0	-9.8			
	10% WB	-16.2	-10.8	-2.1	3.6	8.7	15.4	17.4	16.2	10.7	2.7	-4.6	-12.2			
	10% MCDB	-15.7	-9.4	1.1	9.7	16.5	22.5	23.9	21.3	15.9	6.5	-3.0	-11.6			
	MDBR	13.0	17.2	16.1	13.9	15.7	15.6	14.3	13.4	14.0	13.5	13.0	11.6			
	5% DB	12.8	18.7	18.1	20.7	23.1	20.7	19.4	18.7	20.9	19.0	13.8	10.4			
(36) Mean Daily Temperature Range	5% DB	12.0	16.5	12.2	11.5	11.7	9.0	8.1	9.0	11.5	11.9	11.1	9.6			
	5% WB	12.7	18.4	16.9	18.9	20.5	15.9	15.1	14.4	17.9	17.3	13.2	10.4			
	MCWBR	11.9	16.1	11.5	10.9	10.5	7.4	6.9	7.4	10.0	11.0	10.8	9.6			
	MCWBR	11.9	16.1	11.5	10.9	10.5	7.4	6.9	7.4	10.0	11.0	10.8	9.6			
(40) Clear-Sky Solar Irradiance	taub	0.263	0.286	0.319	0.373	0.450	0.438	0.411	0.377	0.331	0.303	0.285	0.267			
	taud	2.177	2.162	2.205	2.216	2.013	2.165	2.298	2.376	2.454	2.450	2.289	2.159			
	Ebn at Noon	747	832	873	862	805	820	838	851	860	818	719	668			
	Edh at Noon	71	97	113	127	164	142	123	108	89	73	64	62			
(44) All-Sky Solar Radiation	RadAvg	1.40	2.45	3.85	4.94	5.72	6.05	5.61	4.75	3.67	2.39	1.40	1.03			
	RadStd	0.07	0.11	0.15	0.36	0.37	0.36	0.39	0.22	0.32	0.18	0.11	0.07			

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (9 neighbors)	DBAvg	(d) N/A	(e) N/A	(f) -1.44	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A

Nomenclature: See separate page